SE4031 – Games Development

Assignment 02

“Tunnel Rush” - 2D Interactive Game

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Game concept summary

Tunnel Rush is a 2D interactive game where the player needs to go through tunnels while avoiding the tunnel walls to shoot enemies.

Enemies spawn randomly throughout the UI as well as health packs. If the player got hit by an enemy or a wall, player health decreases. Player can refill health using health packs. When the player shoots enemies, the score increases. Game comes to an end when health is 0, while the goal is to survive as long as possible.

The game was developed using Unity and C# scripting.

Core features

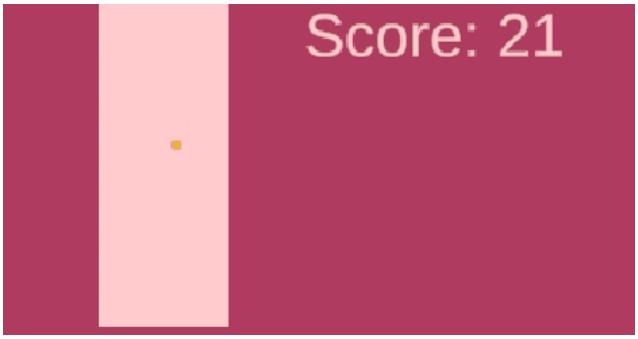
1. Health System

Game starts with player having maximum health value of 100. It displays at the top left corner of the UI. Health decreases when:

- The player collides with tunnel walls.
- The player collides with enemies.

The player stops only when health reaches 0 and health resets to maximum when the game restarts.

The health system is controlled using `PlayerHealth` script. And it includes `maxHealth`, `currentHealth`, A method called `TakeDamage()` and A method called `Heal()`



2. Score System

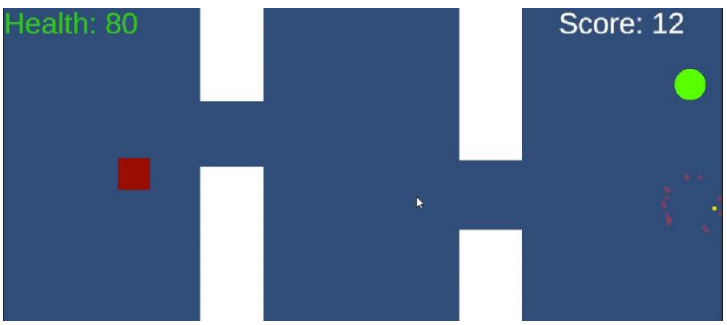
The game includes a scoring system that rewards the player for successfully passing through tunnel gaps and defeating enemies. Score is displayed on the top right of the UI as a number.

Score increases when:

- The player successfully passes through a tunnel hole.
- An enemy is destroyed using a projectile.

Score resets to 0 when the game restarts and increases only once per tunnel.

The score is managed using a `ScoreManager` script. Main functions include, `AddScore(int value)`, `ResetScore()` and `UpdateUI()`



3. Health Packs

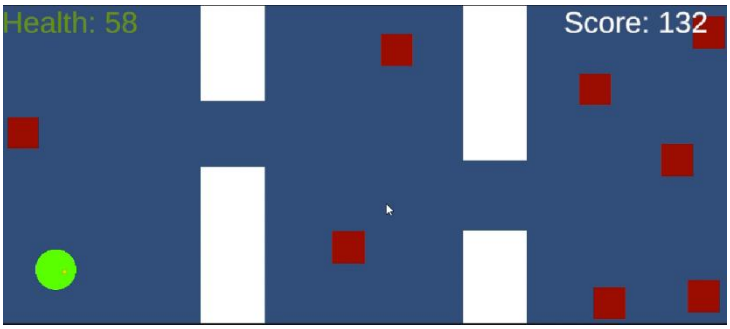
Health packs were added to allow the player to recover health during gameplay.

Features Implemented:

- Health packs are collidable objects.
- They are visually distinct from enemies.
- They are destroyed when collected.
- They increase player health.
- Health does not exceed the maximum value.

When the player collides with a health pack:

- The `Heal()` function in `PlayerHealth` is called.
- The health pack is destroyed.
- Health is increased but capped at max health using `Mathf.Min()`



4. Projectile Attack System

The player has the ability to shoot projectiles and created when left mouse button is clicked.

Projectile is destroyed:

- When it collides with an object.
- After a short duration.

Projectile destroys enemies.

The projectile system includes:

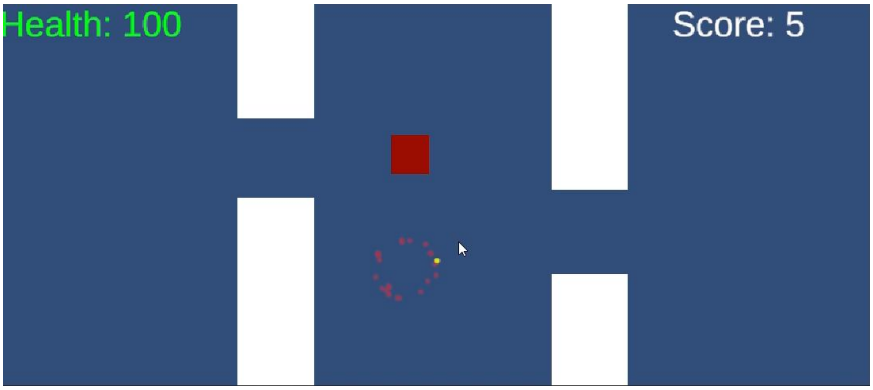
- A shooting script attached to the player.
- A projectile prefab with:
    - Rigidbody2D
    - Collider2D
    - Projectile script

5. Enemy

Features Implemented:

- Enemies are collidable objects.
- Enemies reduce player health on contact.
- Enemies are destroyed when hit by projectiles.
- Destroying enemies increases score.
- Enemies are visually distinct from health packs.

The Enemy script is used for all the above functionalities.



6. Speed Increment Feature

Inside the `PlayerMovement` script, a timer increases using `Time.deltaTime`.

This ensures:

- Early gameplay is easier.
- Game becomes progressively more challenging.

Control Guide

W - player moves up

S – player moves down

D – Player moves right

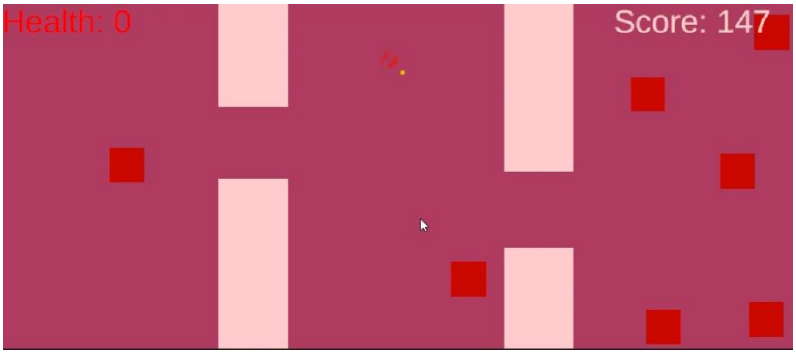
A – Player moves left

R – Restart

Left mouse click – projectile shoot

Creative Features

- Screen shake effect when player takes damage.
- Damage flash effect.
- Enemy death particle effects.
- Clean UI layout using TextMeshPro.
- Range of colors when health decreases.



External assets used:

Audio snippets from Pixabay

Video link: [2D game play recording.zip]